QUANTITATIVE METHODS 1

Worksheet 5 – Time series

Detailed solutions

Exercise 1 – Solution

$$Y_t = T_t + C_t + S_t + R_t,$$
$$Y_t = T_t \times C_t \times S_t \times R_t.$$

Multiplicative seasonality is more plausible when seasonal amplitude grows with the level of the series.

Exercise 2 – Detailed solution

$$MA_{2.5} = 110, \quad MA_{3.5} = 112, \quad MA_{4.5} = 114.5, \quad MA_{5.5} = 117.5, \quad MA_{6.5} = 121.$$
$$CMA_3 = 111, \quad CMA_4 = 113.25, \quad CMA_5 = 116, \quad CMA_6 = 119.25.$$

An even-order moving average lies between two periods, so averaging adjacent moving averages centres the estimate on an observed period.

Exercise 3 – Detailed solution

The trend slope means the underlying level rises by about 6.34 units per quarter.

$$0.789 + 0.903 + 1.052 + 1.255 = 3.999 \approx 4.$$

Trend forecasts are approximately

$$\widehat{T}_{13} = 169.40, \quad \widehat{T}_{14} = 175.74, \quad \widehat{T}_{15} = 182.08, \quad \widehat{T}_{16} = 188.43.$$

Seasonalised forecasts are approximately

$$\widehat{Y}_{13} = 133.66, \quad \widehat{Y}_{14} = 158.68, \quad \widehat{Y}_{15} = 191.55, \quad \widehat{Y}_{16} = 236.48.$$

Using the rounded values reported in the course slides gives about $133.68, 158.68, 191.63, 236.55$.

Multiplicative deseasonalization uses

$$Y_t^{(deseas.)} = \frac{Y_t}{S_j}.$$

Exercise 4 – Detailed solution

Using the course-slide rounded forecasts:

$$\hat{Y} = (133.68, 158.68, 191.63, 236.55).$$

$$e = (2.32, 3.32, -3.63, 8.45).$$
$$MAE = \frac{2.32 + 3.32 + 3.63 + 8.45}{4} = 4.43.$$
$$RMSE = \sqrt{\frac{2.32^2 + 3.32^2 + (-3.63)^2 + 8.45^2}{4}} \approx 5.02.$$

A validation period evaluates the model on observations not used for estimation. A naive forecast provides a simple benchmark that a more complex model should improve upon.